Deep Transformer based Structural Semantic Network for Document Clustering

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***Abstract*—Document clustering is very sensitive to document representation schemes. Traditional document clustering approaches used document content to model features and used representation based on these features thus suffer from problems like vocabulary mismatch and word sense ambiguities, and fail to cater with word-to-word relationship in longer dependency. Consequently, the clustering results lack semantic values. Deep document clustering simultaneously learns semantic repesentation and does clustering using deep neural networks. In this paper, a deep transformer-based structural semantic network for document clustering (DTSN) is proposed, it uses a transformer based approach to learn a document-level semantic representation with long-term text dependency, and exploits graph convolution network in order to capture semantic between documents going layer-by-layer, utilizing reinforcement learning. Extensive series of experiments are performed, and this proposed approach performs substantially better than state-of-the-art deep document clustering approaches. This paper reports a new state-of-the-art on unsupervised text document clustering tasks.**

I. Introduction

Document Clustering is an essential technique in the world of natural language processing which aims to organize vast amounts of documents in distinct groups based on their underlying and inherent similarities. In recent years, deep document clustering has seen quite some interest, as it overcomes some of the difficulties faced by traditional approaches to document clustering.

One such idea was to map data samples to a lower-dimension hidden space in order to extract important features and find data partitions. Most of these approaches also focus only on the internal representations of documents extracted from the semantics learned. However, this proves problematic as documents are limited in their length, as well as express their meaning in various ways. The shortcomings of the majority of these models is the inability to grasp long-term dependency between words, which leads to an inferior understanding of the underlying structure of the document. Alongside this, important information can also be discovered outside the document. Close neighbors of documents that are regarded as the external structural information can also provide vital information that can help in deciding data partitions.

In this paper, a complete end-to-end document clustering model namely DTSN is designed that learns document partition using an ensemble structurally enhanced network. The approach enhances the internal representation of a document using the external representation gathered from its neighbors to calculate an ensemble-enhanced representation which is then used to calculate document partitions. For each document, an encoder-only transformer is used to calculate the internal semantic representation through the usage of Multi-Head Attention, Positional Encoding, and Position-Wise Feed-Forwarding Networks. The common strategy is the usage of Autoencoders(AE) [5] to map the document to a lower dimension; however, it was found that the usage of Transformers and their ability to grasp long-term dependency produced better results. For the external representation of each document, since we wanted a model that provides a consistent and satisfactory result for documents in the form of graphs and the encoding of relevant external information for each data sample, we used graph convolution networks (GCN) [6]. Both models learn different but important semantic information that aids in document clustering. The preferred strategy to improve internal representation with external one is that both approaches run independently and then combine in the end using a normal fusion process with the assumption that the two processes are independent of each other [?], [?], [7], however, the information contained in both processes can help each other produce a more robust representation. As such, a layer by layer enhanced ensemble approach that learns the internal representation as well as the enhanced ensemble representation in every layer by combining the internal and external representations was adapted. The enhanced representation is then used to further reinforce the internal representation of the given document. Furthermore, an objective function for two reasons, to calculate the data partitions and optimize the learning networks of the DTSN model for the document representation.

The Two main contributions of this paper are as follows:

- An end-to-end transformer based deep document clustering model namely DTSN, performs document clustering using a structural enhanced network. It specifically uses internal semantics extracted from a transformer encoder and enhances them using external semantics extracted from a GCN.
- Extensive experiments on real world datasets are carried out against other state-of-the-art clustering datasets to show the effectiveness of the DTSN model. The results to these experiments indicate the substantial improvement DTSN makes in comparison to the other models.

II. Related Works

This paper draws inspiration from three key areas: Transformers, deep clustering based on internal semantics, and deep clustering based on external semantics, while also referencing traditional document clustering methods.

Over the years, Document Clustering has faced many problems, dealing with sparse amount of text in documents, documents expressing the same meaning in different approaches, underlying semantics of the documents, relations between different documents and long term dependencies found in them. Traditionally, numerous approaches have been tried. [20] compares the basic k-means algorithm with bisecting k-means and agglomerative clustering to figure out which is better on numerous datasets. [18] uses Lexical Chains to group related words together alongside WordNet to extract semantics to overcome to limitations of bag-of-words models and capture deeper semantic relationships. [21] leverages frequent word sequences for effective document clustering, it also implements cluster candidate formations and cluster merging. NLP techniques like dependency parsing and named-entity recognition to locate propositions that it uses as the fundamental units of information for clustering.

In recent years, the transformer's self-attention mechanism, that allows the model to give relative importance to certain parts of the data, has increasingly become the focal point of various research studies. To investigates the ability of the self-attention mechanism of BERT in classifying scientific articles and assesses the subset of the most attended words with other feature selection methods in order to identify self-attention as a viable feature selection approach. BERT architecture is further employed by [1] to convert radio-logical reports into embedding vectors to classify these reports by priority. [?] utilizes transformer architecture in natural language processing to predict chemical compound retro-synthesis using text-like embedding of chemical reactions. [2] introduces probabilistic transformers that can transform an arbitrary data embedding into uni-variate Gaussian Mixtures while ensuring low metric distortion. Furthermore, It also presents Contrastive Learning with Bi-directional transformer (CBiT) which enhances sequential recommendation by utilizing bidirectional transformers to process long user sequences sliding window approach creating fine grained division of user sequences.

Likewise, the usage of deep neural networks in clustering has seen a rise as they perform well when it comes to learning feature representations and providing clustering assignments. The Deep clustering network(DCN) uses dimensional reduction techniques using a deep neural network as well as k-means for clustering. Deep embedding clustering(DEC) maps the data space to a lower dimension embedding space which is optimized using KL-divergence-based objective, this was improved by the improved deep embedding clustering(IDEC) [3] which added local structures. Variational deep embedding(VDE) [4] uses the document generation process to understand the document representation which in turn is used for clustering.

To grasp at the underlying semantics in the external representations, the popular approach has been the use of spectral clustering. Which creates a graph with samples as nodes and uses that graph structure as the basis for clustering. Deep neural network for learning graph representations(DNGR) uses spectral clustering with adjacency matrices for similarity matrices. These approaches use nearby graph structures to represent samples, these structures are modeled by deep neural networks. The GCN has also seen great success in this task as many approaches have used it for graph clustering [13]–[15]. The graph autoencoder(GAE) and the variational graph autoencoder(VGAE) have used 2-layer GCNs to learn the semantic representation and then use the autoencoder and the variational autoencoder to reconstruct the adjacency matrix.

There are some models that look to improve the clustering by using both external and internal document information. [15] The structural deep clustering network(SDCN) [15] uses a GCN and an AE and uses a delivery system to reinforce the GCN with the internal representation learned by the AE. Based on this, the attention-driven graph clustering(AGCN) [15] created a system to fuse the two representations by the aggregation of multi-scale features incorporated at various layers. The deep structural enhanced network for document clustering(DSEDC) [19] further improved on the SDCN and introduced the creation of an enhanced ensemble representation by combining the external and internal representation. SDCN and AGCN however fall short due to their focus on the GCN-based structural representation, as the noise for text documents in this case is quite large and effects the results, not to mention the semantics learned for the content are also mostly put to the wayside. The DSEDC model employs the usage of AEs which do a great job at learning internal semantics but largely fail at grasping long term dependencies, something that transformers are designed to do well.

Recent works in 2025 have taken some fresh new approaches. One includes community-detection based initialization followed by merging of communities to perform clustering [12]. Another includes an approach similar to ours, utilizing the encoding of transformers to create embeddings to feed to autoencoders for clustering [15], however, this approach does not take advantage of a GCN model to extract the external structure of the data.

III. The Proposed model: DTSN

In this section, details are shown for the proposed DTSN model by explaining the overall framework followed by individual modules and optimizations.

A. Proposed Framework

Two modules define the framework presented for the proposed model, the GCN Module and the Transformer Module. For every document sample that is input as data, the GCN module creates a reasonable data partition using its closeness. There are multiple layers of the GCN model which outputs an external data representation.

The internal and ensemble representation for the document is calculated by the Transformer module while keeping in mind both the internal and external representations. In the first layer, it uses the internal representation it learned with the external representation learned by the GCN to create the ensemble document representation. The internal document representation is further improved through reinforcement learning by using the ensemble representation as the input for the next layer to learn the internal representation so in the I-th layer, the internal representation is used to create the ensemble representation which is used as input for the internal representation in the I+1-th layer.

The ensemble representation is used as input for the Document Clustering module alongside the external representation from the GCN module, this module uses them to create document partitions as well as using a joint objective function to further fine tune the results alongside the representations.

B. Transformer Encoder Module

Transformer network architecture is a novel way to tackle sequence to sequence modeling tasks. Its self-attention mechanism excels at capturing dependencies within the data regardless of whether those connections happen over time (temporal) or space (spatial); allowing significantly more parallelization than its counterparts. In this study, the encoder layer of the transformer network is investigated to capture and produce context aware internal structural embeddings for each document sample.

In order for the transformer's encoder layer to make use of order of sequence, information about the relative and absolute positions of the input data is injected into the input sequence. the sine and cosine functions have been used to generate the positional encodings where p is the position and k is the dimension:

$$\mathrm{PE}(p, 2k) = \sin\left(\frac{\mathrm{p}}{1000^{2k/d_{\mathrm{model}}}}\right)$$

$$\mathrm{PE}(p, 2k+1) = \cos\left(\frac{\mathrm{p}}{1000^{2k/d_{\mathrm{model}}}}\right)$$

The input sequence is improved by adding the positional encoding to it and it is then sent to the encoder stack.

A transformer encoder stack can contain identical layers up to N which contain 2 further sublayers. Multi-head self-attentiona mechanisms are in the first layer while the second consists of fully connected feed-forward networks optimized for position. Residual connections are employed around each sublayer, followed by a normalizing layer.

The multi-attention layer receives the input and projects it down into key, query, and value vectors. The model is able to combine information from different representations at different positions using multi-head attention, it then also computes using this information. The key, value, and query vectors are linearly projected h times using distinct learned linear projections, and the attention function is applied in parallel to each of these projections.

Query vectors are mapped to a set of key-value pairs which are in turn mapped to an output vector using an attention function. Vecotrs that are weighted sums of values are used to create the output, a compatibility function is used for each query with corresponding key

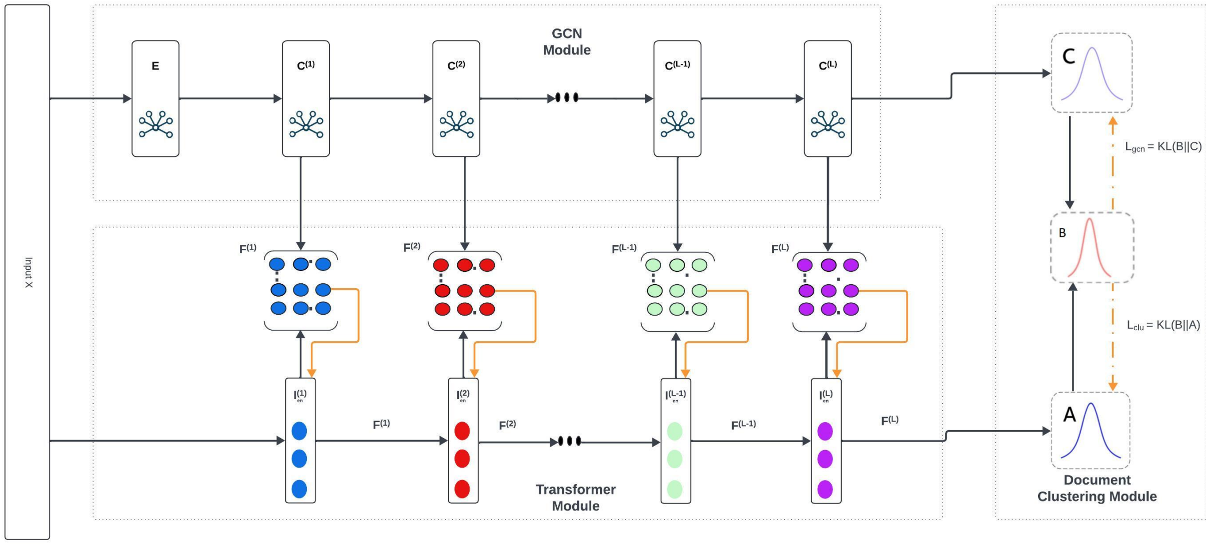

Fig. 1. The DTSN model

values to calculate the weights. Attention function on the set of queries is computed simultaneously along with key, values, and dimension of key vector dk as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{\text{dk}}}\right)V$$

Final representation is the concatenation of each output from the attention function. The output from the multi-head attention layer is then normalized before sending it to the position-wise feed forward network where a fully connected feed-forward network is applied on each position separately and identically but having different parameters from layer to layer. This contains 2 linear transformations with a ReLU activation in between.

C. GCN Module

In the GCN module, Graph Convolutional Networks (GCNs) have a crucial role in capturing external structural representations of each document sample. Each document representation C(l) at l-th layer is calculated through the following convolution calculation:

$$C^{(l)} = f(C^{(l-1)}, E|W_c^{(l-1)})$$
$$f(C^{(l-1)}, E|W_c^{(l-1)}) = \Phi_c(\overline{D}^{-\frac{1}{2}}\overline{ED}^{-\frac{1}{2}}Z^{(l)}W_c^{(l)})$$
$$\overline{E} = \overline{E} + I$$
$$\overline{D}_{ii} = \sum_{j=1}^{N} \overline{E}_{ij}$$

where the GCN structure of each data sample is represented by E. D_{ii} is the degree matrix of of the GCN structure and W_c^l is the weight matrix at the GCN's l-th layer . Various strategies have been utilized to assess the similarity for each document sample, both with or without explicit graph structure. For samples having clear graph structures, GCNs have been made using the respective graph structure. For the remaining samples, not having defined graph representation, KNN model has been used to determine K-nearest neighbors and their similarities are leveraged to construct the GCN structure.

D. DC Module

The DC module is subdivided into two tasks. First, the enhanced representation that is generated using the transformer module are used to learn the document partitions. Secondly, the networks that create the representation for the documents are optimized. An objective function is being employed which makes use of confidence of clustering partitioning to achieve both tasks all together. For learning document partition, to measure the similarity between the document representations and cluster centers, a kernel in the form of Student's-distrubtion has been used.Student's-distribution has been used. The clustering distribution A and the targeting distribution B are being calculated as follows:

$$a_{ij} = \frac{(1 + ||f_i - \mu_j||^2/\alpha)^{\frac{-\alpha+1}{2}}}{\sum_{j'=1}^{J}(1 + ||f_i - \mu_{j'}||^2/\alpha)^{\frac{-\alpha+1}{2}}}$$
$$b_{ij} = \frac{a_{ij}^2/\tau_j}{\sum_{j'=1}^{J} a_{ij}^2/\tau_j}$$

where soft cluster frequency is τ_j, Student's distribution degree of freedom is represented byα, f_i is the ensemble document representation and μ_j is the cluster center. Using these distributions the derivation of the objective function used is as follows:

$$L_{clu} = KL(B||A) = \sum_{i=1}^{N}\sum_{j=1}^{J} p_{ij} log \frac{b_{ij}}{a_{ij}}$$

where number of documents and clusters are assigned N and J respectively.

This paper also uses the same method for optimization of the GCN module by calculating similarity between external representation c_i and μ_j, the distribution is as follows:

$$c_i j = \frac{(1 + ||c_i - \mu_j||^2/\alpha)^{\frac{-\alpha+1}{2}}}{\sum_{j'=1}^{J}(1 + ||c_i - \mu_{j'}||^2/\alpha)^{\frac{-\alpha+1}{2}}}$$

where distribution of the clustering result is C which is created by the C^L layer. Similarly, the targeted distribution P is used to guide the GCN with the following objective function.

$$L_{gcn} = KL(B||C) = \sum_{i=1}^{N}\sum_{j=1}^{J} b_{ij} log \frac{b_{ij}}{c_{ij}}$$

With the final loss being as follows:

$$L = L_{clu} + w_g L_{gcn}$$

Optimization is achieved through tuning the document representational networks. The objective function combines clustering loss, and GCN tuning loss to improve document clustering accuracy and representational learning. The Ranger optimizer [**?**] is also used that combines the RAdam optimizer [16] and the LookAhead optimizer [17] to optimize cluster centers and network parameters.

IV. Experiments

A. Datasets

Two real-world datasets will be used in experimentations, whose descriptions are shown below:

- ACM[1]: From the ACM website, it is a popular dataset of documents that contains links. Keywords are reperesented by features which are bag of words as well as lusts of coauthor links between documents. 3 Classes are available for papers printed in this dataset, these include database, wireless communication and data mining.
- BBC[2]: BBC is a web text document dataset with 5 classes of news, business, entertainment, politics, sports and tech. It consists of 2225 documents from the BBC website containing news stories from 2004-2005 This dataset contains 5 classes related to news. All of the data from this dataset are web text documents. The news classes include business, entertainment, politics, sports and tech. The total documents in it are 2225 which include BBC website news stories from 2004-2005.

B. Baseline Methods

- Internal representation methods:
  K-means [**?**] is the most popular clustering model using traditional techniques and also does not use deep learning.
  Lexical Chains [18] are made using semantics from WordNet and are used as features for document clustering
  AE [5] and VAE [**?**] use autoencoders and variational autoencoders respectively to learn representations for clustering
  DEC [**?**] formulates an objective for clustering whose aim it is to direct the learning of data representations.
  IDEC [3] adds reconstruction loss to improve on DEC.
- internal representation methods:
  Spectral [**?**] uses weighted graphs by using the samples as nodes and further uses the structure of said graphs for clustering.
  DNGR [**?**] is similar to spectral but it uses adjacency matrix in place of similarity matrix instead.
- Internal and external representation methods:
  GAE and VGAE [**?**] use GCNs to create representations for variational and normal autoencoders.
  ARGA and ARVGA [13] use adversarial models to improve on the previous two models.
  SDCN [**?**] merges GCN and a DNN to create a network for improving clustering.
  SDCNQ [**?**] variant on previous model using a different distribution Q.
  AGCN [15] created a system to fuse the two representations by the aggregation of multi-scale features embedded at different layers,
  DSEDC [19] is a enhanced clustering network that combines a GCN and AE to create an enhanced representation to construct a clustering network
  DSEDC-S3 [19] is a variation of the DSEDC model where the 3 layers are enhanced from the beginning layer

C. Experimental Setup

For the Baseline methods with pretraining processes(AE, DEC, IDEC, SDCN, AGCN, DSEDC), the same dimensions are used: i-500-500-200-10 where i is the input data dimension. They're trained for 30 epochs with 10^{-3} learning rate by the Adam algorithm. For included baseline methods, reported results from given papers are used. For this paper's model, the dimensions of the GCN and Transformer are all equal to the input

[1]http://dl.acm.org/
[2]http://mlg.ucd.ie/datasets/bbc.html

data's dimensions with 5 layers, with K = 30 on non-graph datasets as well as α = 2, γ = 0.5 and w_g = 0.01. For the ACM dataset, Positional Encoding in the transformer is used while in the bbc dataset it is not. the DTSN model is trained for 300 epochs with 10^{-4} learning rate using the Ranger algorithm.

D. Experiment Results

Here, the results of the proposed model are provided on the datasets used and compared to baseline models. Table 1 shows these results using the two datasets, ACM contains links while BBC doesn't.

As can be observed, in all three metrics, this paper's DTSN model outperforms the other baseline models by a wide margin, these results provide substantial evidence to advocate for the proposed DTSN model. Discussing the results, the model improves on the DSEDC model by 11.59%, 27.2% and 30.12% on the BBC dataset with respect to Accuracy, NMI and ARI. It also improves on the DSEDC-S3 model by 7.88%, 32.84% and 25.62% on the ACM dataset with respect to Accuracy, NMI and ARI.

TABLE I
REAL-WORLD DATASETS CLUSTERING RESULTS

Methods	BBC			ACM		
	ACC	NMI	ARI	ACC	NMI	ARI
Kmeans	51.15	32.97	14.37	67.31	32.44	30.60
AE	51.73	51.33	37.33	81.83	49.30	54.64
VAE	70.23	53.04	43.04	69.75	31.16	32.53
DEC	77.06	61.51	59.59	84.33	54.54	60.64
IDEC	76.40	64.48	61.72	85.12	56.61	62.16
Spectral	71.96	61.68	43.73	73.62	40.63	40.89
DNGR	76.90	56.21	50.16	37.69	1.76	1.63
GAE	79.75	65.54	61.69	84.52	55.38	59.46
VGAE	79.91	63.43	63.23	84.13	53.20	57.72
ARGA	84.31	61.56	65.16	83.50	51.09	56.04
ARVGA	85.44	64.73	66.75	85.45	54.86	61.22
SDCN_Q	77.02	62.97	60.91	86.95	58.90	65.25
SDCN	77.55	65.28	62.58	90.45	68.31	73.91
AGCN	72.60	62.98	57.38	90.59	68.38	74.20
DSEDC	88.61	75.35	74.78	91.33	70.99	76.14
DSEDC-S3	88.16	72.25	71.25	92.32	73.72	78.66
DTSN	**98.88**	**95.85**	**97.31**	**99.60**	**97.93**	**98.81**

V. CONCLUSION

This study introduced DTSN, a novel end-to-end deep document clustering framework that integrates an ensemble reinforcement-based mechanism and transformer architectures to enhance both internal and external semantic representations. A joint optimization objective facilitates simultaneous improvement in clustering quality and representation learning. Extensive evaluations on real-world datasets demonstrate its efficacy. Future work may focus on extending the model to diverse datasets and enhancing performance metrics such as NMI and ARI.

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